Pre-Analysis-Plan for Collischon/Schwarz - #ManyDaughters

Hypothesis 1

Having daughters leads to stronger support for gender equality (as measured by attitudes).

> That is, denoting X as a measure of support for gender equality, (X for respondents with daughters) > (X for respondents without daughters).

Variables:

> Outcome: plh0308_v2 -> GE
>
> (values taken as is)
>
> Regressor of interest: kidsex01 -> daughter (β_1)
>
> (daughter=1 if kidsex01==2; daughter=0 if kidsex01==1)
>
> Controls: year of birth (ple0010_h) as indicators -> birthyear
>
> (values taken as is)
>
> Gender (sex) -> male
>
> (male=1 if sex==1; male=0 if sex==2)
>
> Child year of birth (kidgeb01) as indicators -> birthyear_child
>
> (values taken as is)
>
> Marital status (pgfamstd) as indicators -> famstat
>
> (values taken as is)
>
> Years of education (pgbilzeit) continuous -> yeduc
>
> (values taken as is)

Inclusion/exclusion criteria:

- Individuals with one child, dropping observations after having multiple children (keep if kidsex02==-2)
- Only survey year 2018 is used as only it contains the outcome measure
- We drop all cases with missing information in any of the variables used in the regression (values <0 in all variables in the data)

Regression equation:

$$GE_i = \beta_1 daughter_i + \beta_2 X_i + \epsilon_i$$

With GE being the aforementioned measure on gender quality, *daughter* is an indicator variable indicating whether one's first child is female (reference: sons), *X* is a set of control variables (described above) and ϵ is an idiosyncratic error term.

We use heteroscedasticity-robust standard errors in our analysis; the model is estimated via OLS.

Interpretation:

- We use the coefficient direction and p-value corresponding to the estimate of $\beta_1$to test the hypothesis. If the estimate is positive and the p-values is below 0.05, we do not reject the hypothesis that having daughters leads to more support for gender equality.

Hypothesis 2

Having daughters leads to more left-leaning political preferences.

That is, denoting Y as a measure of left-leaning political preferences, (Y for respondents with daughters) > (Y for respondents without daughters).

Variables:

Outcome: plh0004 -> PP

(values taken as is)

Regressor of interest: kidsex01 -> daughter (β_1)

(daughter=1 if kidsex01==2; daughter=0 if kidsex01==1)

Controls: year of birth (ple0010_h) as indicators -> birthyear

(values taken as is)

Gender (sex) -> male

(male=1 if sex==1; male=0 if sex==2)

Child year of birth (kidgeb01) as indicators -> birthyear_child

(values taken as is)

Marital status (pgfamstd) as indicators -> famstat

(values taken as is)

Years of education (pgbilzeit) continuous -> yeduc

Inclusion/exclusion criteria:

- Individuals with one child, dropping observations after having multiple children (keep if kidsex02==-2)
- Only survey year 2019
- We drop all cases with missing information in any of the variables used in the regression (values <0 in all variables in the data)

Regression equation:

$$PP_i = \beta_1 daughter_i + \beta_2 X_i + \epsilon_i$$

With PP being the left-right scale, *daughter* is an indicator variable indicating whether one's first child is female (reference: sons), *X* is a set of control variables (described above) and ϵ is an idiosyncratic error term.

We use heteroscedasticity-robust standard errors in our analysis; the model is estimated via OLS.

Interpretation:

- We use the coefficient direction and p-value corresponding to the estimate of $\beta_1$to test the hypothesis. If the estimate is negative (i.e. less right-leaning) and the p-values is below 0.05, we do not reject the hypothesis that having daughters leads to more left-leaning political preferences, as compared to having sons

Hypothesis 3:

Having daughters leads to more gender equality in the household (as measured by behaviors or outcomes).

That is, denoting Z as a measure of gender equality in the household, (Z for respondents with daughters) > (Z for respondents without daughters)

Variables:

Outcome: Absolute difference in actual working hours between partners (DWH)

(plb0186_h -> hours_week for respondent and partner; values taken as is; "-2" for non-working is coded as "0"; DWH is the absolute difference between these values)

Regressor of interest: kidsex01 -> daughter (β_1)

(daughter=1 if kidsex01==2; daughter=0 if kidsex01==1)

Controls: year of birth (ple0010_h) as indicators -> birthyear

(values taken as is)

Gender (sex) -> male

(male=1 if sex==1; male=0 if sex==2)

Child year of birth (kidgeb01) as indicators -> birthyear_child

(values taken as is)

Marital status (pgfamstd) as indicators -> famstat

(values taken as is)

Years of education (pgbilzeit) continuous -> yeduc

(values taken as is)

Survey year (syear)

(values taken as is)

Inclusion/exclusion criteria:

- Individuals with one child, dropping observations after having multiple children (keep if kidsex02==-2)
- We drop all cases with missing information in any of the variables used in the regression
- Only include households with partners (e.g. no single parents) and children; hgtyp1hh=4/5/6
- Only heterosexual couples (drop if gender of the partner equal respondent's gender)
- We use all survey years available

Regression equation:

$$DWH_{it} = \beta_1 daughter_{it} + \beta_2 X_{it} + \lambda syear_t + \epsilon_i$$

With DWH being the difference in actual working hours within the couple, *daughter* is an indicator variable indicating whether one's first child is female (reference: sons), *X* is a set of control variables (described above) and ϵ is an idiosyncratic error term.

We use clustered standard errors on the household level in our analysis; the model is estimated via OLS.

Interpretation:

- We use the coefficient direction and p-value corresponding to the estimate of $\beta_1$to test the hypothesis. If the estimate is negative (i.e. smaller differences in working hours) and the p-values is below 0.05, we do not reject the hypothesis that having daughters leads to more gender equal behavior.

Hypothesis 4

Having daughters leads to a larger increase in the support of gender equality for men than for women (with gender equality measured by attitudes).

> That is, denoting X as a measure of support for gender equality, X for male respondents with daughters − X for male respondents without daughters! > X for female respondents with daughters − X for female respondents without daughters! .

Variables:

Outcome: plh0308_v2 -> GE

(values taken as is)

Regressor of interest: male*daughter (α_1)

(see variable definitions below)

Controls: year of birth (ple0010_h) as indicators -> birthyear

(values taken as is)

Gender (sex) -> male

(male=1 if sex==1; male=0 if sex==2)

Child year of birth (kidgeb01) as indicators -> birthyear_child

(values taken as is)

Marital status (pgfamstd) as indicators -> famstat

(values taken as is)

Years of education (pgbilzeit) continuous -> yeduc

(values taken as is)

kidsex01 -> daughter (β_1)

(daughter=1 if kidsex01==2; daughter=0 if kidsex01==1)

Inclusion/exclusion criteria:

- Individuals with one child, dropping observations after having multiple children (keep if kidsex02==-2)
- Only survey year 2018 is used as only it contains the outcome measure
- We drop all cases with missing information in any of the variables used in the regression

Regression equation:

$$GE_i = \beta_1 daughter_i + \alpha_1 male_i \cdot daughter_i + \beta_2 X_i + \epsilon_i$$

With GE being the aforementioned measure on gender quality, *daughter* is an indicator variable indicating whether one's first child is female (reference: sons), *X* is a set of control variables (described above) and ϵ is an idiosyncratic error term. α_1 is the interaction effect between being male and having a daughter.

We use heteroscedasticity-robust standard errors in our analysis; the model is estimated via OLS.

Interpretation:

- We use the coefficient direction and p-value corresponding to the estimate of $\alpha_1$to test the hypothesis. If the estimate is positive and the p-values is below 0.05, we do not reject the hypothesis

that having daughters leads to a larger increase in the support of gender equality for men than for women, under the assumption that β_1 is positive and significant (i.e. that there is no negative or zero effect of having daughters for women).